Lecture 16 Scribe

L15_L17.pdf
PDF

Generate a formal lecture scribe for Lecture 16 that serves as a high-fidelity, exam-ready reconstruction of the attached lecture slides . You must strictly adhere to the provided materials, ensuring that no content, intuition, or information is included unless it is explicitly traceable to the attached context. If a theorem or result is stated without a proof in the provided materials, you must not include or search for a proof; do not assume knowledge from previous lectures or external sources. The scribe must contain Definitions and notation, Assumptions and conditions, Statements of theorems or results, Proofs or proof sketches, step by step, Worked examples, with intermediate steps, Clear logical flow between ideas if present in the lecture slide.The scribe must be organized into a logical, hierarchical structure using sections and headings, documenting all definitions, notations, assumptions, and results exactly as presented. Every worked example must include the specific intermediate steps provided in the text. Your output must prioritize technical precision and strict faithfulness to the PDF, excluding any extra explanations, tutorials, or "common knowledge" not found within the document.

Below is a faithful reconstruction from the attached combined slide deck **“Lecture L15_16_17_18: Joint Random Variables + Joint Distributions + Examples.”** The file does **not** explicitly mark a standalone “Lecture 16” boundary inside the deck, so this scribe only documents material that is explicitly present in the attached slides and does not infer any undocumented split between lectures.

Lecture Scribe: Joint Random Variables, Joint Distributions, and Joint CDF

1. Lecture Metadata

Course: CSE400 - Fundamentals of Probability in Computing
Lecture Deck Title: Lecture L15_16_17_18: Joint Random Variables + Joint Distributions + Examples
Instructor: Dhaval Patel, PhD
Date: March 10, 2026

2. Outline from the Slides

The slide deck presents the following flow:

1. Why Study Joint Random Variables?
2. Discrete Case
   - Joint PMF Table Representation
   - Example: Rolling Two Dice
3. Continuous Case
   - Geometric Interpretation
   - Worked Example
4. Joint Cumulative Distribution Function
   - Definition of Joint CDF
   - Continuous Case
   - Discrete Case
   - Properties of the Joint CDF
5. Comprehensive Example

3. Why Study Joint Random Variables?

The slides state that in science and in real life, we are often interested in **two (or more) random variables at the same time**. Examples listed in the slides are:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

The slides then ask:

Think: What kind of relationship would you expect in each case? Why?

3.1 Application Examples Given in the Slides

Example 1: Smart Transportation
Two random variables are introduced:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question posed on the slide: Can speed be high when the traffic density is also high?
The slide records the answer format as: **Yes / No**.

The slide then states: **So, X and Y are correlated random variables.**

Applications listed:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance
Two random variables are introduced:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

The slide states that throughput depends on SNR, and that both vary due to fading, interference, and congestion.

Question posed on the slide: If SNR is high, will throughput always be high?
The slide records the answer format as: **Yes / No**.

The slide adds:

- Reasoning: Network Congestion (condition relationships)
- Applications: 5G scheduling, adaptive modulation, network planning

Example 3: Weather Prediction

- $X =$ Rainfall
- $Y =$ Temperature

Application: weather forecasting.

Example 4: Drone Delivery

- $X =$ Wind Speed
- $Y =$ Delivery Time

Applications: logistics and UAV systems.

Example 5: Rolling Two Dice
The slide states:

- Roll two dice $(X \text{ and } Y)$
- Consider conditions such as $X + Y = 7$ and $X > Y$
- This requires a joint PMF.

3.2 Purpose of Joint Distributions

The slides state that in such situations the random variables have a **joint distribution** that allows us to:

- compute probabilities of events involving both variables, and
- understand the **relationship** between the variables.

The slides further state:

- This is simplest when the variables are **independent**.
- When they are not, we use **covariance** and **correlation** as measures of the nature of the dependence between them.

4. Discrete Case: Joint PMF

4.1 Setup and Notation

Suppose X and Y are discrete random variables with:

$$X \in \{x_1, x_2, \ldots, x_n\}, \qquad Y \in \{y_1, y_2, \ldots, y_m\}.$$

Then the ordered pair (X, Y) takes values in the product set:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}.$$

4.2 Definition (Joint PMF)

The slides define the **joint PMF** by

$$p(x_i, y_j) = P(X = x_i,\ Y = y_j).$$

The slide states that this gives the probability of the joint outcome occurring simultaneously.

4.3 Table Representation

The slides organize the joint PMF in a table with rows indexed by $x_1, \ldots, x_n$ and columns indexed by $y_1, \ldots, y_m$, where each cell contains $p(x_i, y_j)$.

4.4 Properties of the Joint PMF

The slides list these properties:

$$0 \leq p(x_i, y_j) \leq 1$$

and

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1.$$

5. Worked Example: Rolling Two Dice

5.1 Experiment and Variables

The experiment is:

- Roll two fair dice.

The slides define:

- $X =$ value on the first die
- $T =$ total (sum) of both dice

Since each outcome (i, j) has probability

$$\frac{1}{36},$$

the slides construct the joint probability table of (X, T).

5.2 Joint Probability Table of (X, T)

The table shown on the slide is:

$X \backslash T$	2	3	4	5	6	7	8	9	10	11	12
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0	0
2	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0
3	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0
4	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0
5	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0
6	0	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

5.3 Observations Stated in the Slides

The slide explicitly records:

- Entries are either 0 or $\frac{1}{36}$
- X and T are **not independent**.

6. Continuous Case: Joint PDF

6.1 Analogy with the Discrete Case

The slides state that the continuous case is analogous to the discrete case:

- Discrete values $\longrightarrow$ Continuous intervals
- Joint PMF $\longrightarrow$ Joint PDF
- Sums $\longrightarrow$ Double integrals

6.2 Product Region

If

$$X \in [a, b], \qquad Y \in [c, d],$$

then the pair (X, Y) takes values in the product region

$$[a, b] \times [c, d].$$

6.3 Definition (Joint PDF)

The slides define a function $f(x, y)$ to be the **joint probability density function** of X and Y if

$$P\big((X, Y) \text{ lies in a small rectangle of area } dx\, dy\big) \approx f(x, y)\, dx\, dy.$$

6.4 Geometric Interpretation

The slides illustrate a rectangle inside the region $[a, b] \times [c, d]$, labeling the small rectangle with side lengths dx and dy, and assigning probability

$$\text{Prob.} = f(x, y)\, dx\, dy.$$

6.5 Properties of the Joint PDF

The slides list:

$$f(x, y) \geq 0$$

and

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1.$$

The slides also state:

The joint PDF $f(x, y)$ may take values greater than 1. It is a probability density, not a probability.

7. Worked Example in the Continuous Case

7.1 Given

The slides give:

$$f(x, y) = 4xy, \qquad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

The slides also state that jointly (X, Y) takes values in the **unit square** $[0, 1]^2$.

7.2 Tasks Listed in the Slides

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize the event

   $A = \{X < 0.5,\ Y > 0.5\}.$
3. Compute $P(A)$.

7.3 Visualization of the Event

The slide shades the region

$$A = \{X < 0.5,\ Y > 0.5\}$$

inside the unit square $[0, 1]^2$. It is the upper-left rectangle cut out by the lines $x = 0.5$ and $y = 0.5$.

7.4 Example Solution: Step 1 — Check Validity (Normalization)

The slides show the normalization calculation step by step:

$$\int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 \left(\int_0^1 4xy\,dx \right) dy = \int_0^1 2y\,dy = [y^2]_0^1 = 1.$$

The slide then concludes:

$$f(x, y) \geq 0$$

and total probability is 1, so it is a **valid joint PDF**.

7.5 Example Solution: Step 2 — Compute $P(A)$

The slides compute:

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx = \int_0^{0.5} [2xy^2]_{0.5}^1\,dx = \int_0^{0.5} 2x(1 - 0.25)\,dx = \int_0^{0.5} \frac{3}{2}x\,dx =$$

8. Joint Cumulative Distribution Function (Joint CDF)

8.1 Event Notation

Suppose X and Y are jointly distributed random variables.

The slides use the notation

$$X \leq x, \quad Y \leq y$$

to mean the event

$$\{X \leq x \text{ and } Y \leq y\}.$$

8.2 Definition (Joint CDF)

The slides define the joint cumulative distribution function as

$$F(x, y) = P(X \leq x,\ Y \leq y).$$

9. Joint CDF — Continuous Case

If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, then the slides state that the joint CDF is obtained by integrating the density:

$$F(x, y) = \int_c^y \int_a^x f(u, v)\,du\,dv.$$

9.1 Recovering the Joint PDF

The slides state that since there are two variables, we use partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x \, \partial y} F(x, y).$$

10. Joint CDF — Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, then the slides state that the joint CDF is obtained by summing probabilities:

$$F(x, y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j).$$

10.1 Interpretation Given in the Slides

The slides explain this as:

- Add all probabilities in the table
- For rows with $x_i \le x$
- And columns with $y_j \le y$.

11. Properties of the Joint CDF

The slides state that the joint CDF $F(x, y)$ of X and Y must satisfy several properties.

Property 1: Non-decreasing

$$F(x, y) \text{ is non-decreasing.}$$

If x or y increase, then $F(x, y)$ must stay constant or increase.

Property 2: Zero at the lower-left of the joint range

$$F(x, y) = 0 \text{ at the lower-left of the joint range.}$$

If the lower-left corner is $(-\infty, -\infty)$, then

$$\lim_{(x,y) \to (-\infty, -\infty)} F(x, y) = 0.$$

Property 3: One at the upper-right of the joint range

$$F(x, y) = 1 \text{ at the upper-right of the joint range.}$$

If the upper-right corner is (∞, ∞), then

$$\lim_{(x,y)\to(\infty,\infty)} F(x,y) = 1.$$

12. Notes on Faithfulness to the Slides

- The attached deck is a **combined** lecture deck, not a separately labeled standalone Lecture 16 file.
- The slides explicitly include definitions, formulas, examples, and the worked continuous example above.
- No external theorem proofs or additional explanations have been added.
- Where the slides only state a formula or property, the scribe only states that formula or property.
- Where the slides include intermediate steps, those steps have been reproduced.

If you want, I can now convert this into a cleaner **LaTeX scribe format** without changing the content.